MOS TECHNOLOGY, INC.

Valley Forge Corporate Center
950 Rittenhouse Road
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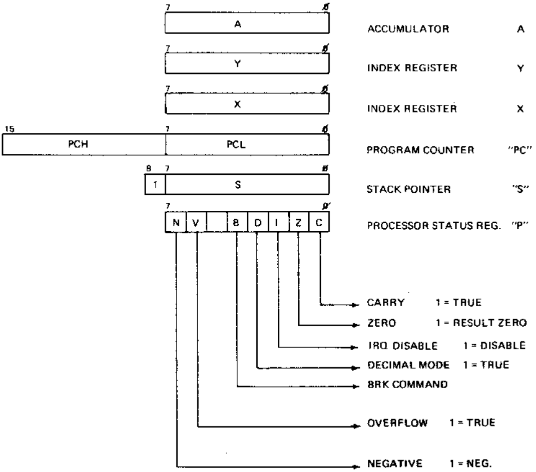
MCS6500 INSTRUCTION SET SUMMARY

PROCESSOR PROGRAMMING MODEL



INSTRUCTIONS		IMMEDIATE			ABSOLUTE			ZERO PAGE			ACCUM.			IMPLIED			(IND,X)			(IND),Y			Z,PAGE,X			ABS,X			ABS,Y			RELATIVE			INDIRECT			Z,PAGE,Y			CONDITION CODES					
MNEMONIC	OPERATION	OP	N	#	OP	N	#	OP	N	#	OP	N	#	OP	N	#	OP	N	#	OP	N	#	OP	N	#	OP	N	#	OP	N	#	OP	N	#	OP	N	#	OP	N	#	N	Z	C	I	D	V
ADC	A+M+C → A (4) (1)	69	2	2	6D	4	3	65	3	2							61	6	2	71	5	2	75	4	2	7D	4	3	79	4	3										✓	✓	✓	-	-	✓
AND	A∧M → A (1)	29	2	2	2D	4	3	25	3	2							21	6	2	31	5	2	35	4	2	3D	4	3	39	4	3										✓	✓	-	-	-	-
ASL	C ← [7 … 0] ← 0				0E	6	3	06	5	2	0A	2	1										16	6	2	1E	7	3													✓	✓	✓	-	-	-
BCC	BRANCH ON C=0 (2)																															90	2	2							-	-	-	-	-	-
BCS	BRANCH ON C=1 (2)																															B0	2	2							-	-	-	-	-	-
BEQ	BRANCH ON Z=1 (2)																															F0	2	2							-	-	-	-	-	-
BIT	A∧M				2C	4	3	24	3	2																															M_7	✓	-	-	-	M_6
BMI	BRANCH ON N=1 (2)																															30	2	2							-	-	-	-	-	-
BNE	BRANCH ON Z=0 (2)																															D0	2	2							-	-	-	-	-	-
BPL	BRANCH ON N=0 (2)																															10	2	2							-	-	-	-	-	-
BRK	(See Fig. 1)													00	7	1																									-	-	-	1	-	-
BVC	BRANCH ON V=0 (2)																															50	2	2							-	-	-	-	-	-
BVS	BRANCH ON V=1 (2)																															70	2	2							-	-	-	-	-	-
CLC	0 → C													18	2	1																									-	-	0	-	-	-
CLD	0 → D													D8	2	1																									-	-	-	-	0	-
CLI	0 → I													58	2	1																									-	-	-	0	-	-
CLV	0 → V													B8	2	1																									-	-	-	-	-	0
CMP	A-M (1)	C9	2	2	CD	4	3	C5	3	2							C1	6	2	D1	5	2	D5	4	2	DD	4	3	D9	4	3										✓	✓	✓	-	-	-
CPX	X-M	E0	2	2	EC	4	3	E4	3	2																															✓	✓	✓	-	-	-
CPY	Y-M	C0	2	2	CC	4	3	C4	3	2																															✓	✓	✓	-	-	-
DEC	M-1 → M				CE	6	3	C6	5	2													D6	6	2	DE	7	3													✓	✓	-	-	-	-
DEX	X-1 → X													CA	2	1																									✓	✓	-	-	-	-
DEY	Y-1 → Y													88	2	1																									✓	✓	-	-	-	-
EOR	A⊻M → A (1)	49	2	2	4D	4	3	45	3	2							41	6	2	51	5	2	55	4	2	5D	4	3	59	4	3										✓	✓	-	-	-	-
INC	M+1 → M				EE	6	3	E6	5	2													F6	6	2	FE	7	3													✓	✓	-	-	-	-
INX	X+1 → X													E8	2	1																									✓	✓	-	-	-	-
INY	Y+1 → Y													C8	2	1																									✓	✓	-	-	-	-
JMP	JUMP TO NEW LOC.				4C	3	3																												6C	5	3				-	-	-	-	-	-
JSR	(See Fig. 2) JUMP SUB				20	6	3																																		-	-	-	-	-	-
LDA	M → A (1)	A9	2	2	AD	4	3	A5	3	2							A1	6	2	B1	5	2	B5	4	2	BD	4	3	B9	4	3										✓	✓	-	-	-	-

		IMMEDIATE			ABSOLUTE			ZERO PAGE			ACCUM.			IMPLIED			(IND,X)			(IND),Y			Z,PAGE,X			ABS,X			ABS,Y			RELATIVE			INDIRECT			Z,PAGE,Y			CONDITION CODES					
MNEMONIC	OPERATION	OP	N	#	OP	N	#	OP	N	#	OP	N	#	OP	N	#	OP	N	#	OP	N	#	OP	N	#	OP	N	#	OP	N	#	OP	N	#	OP	N	#	OP	N	#	N	Z	C	I	D	V
LDX	M → X (1)	A2	2	2	AE	4	3	A6	3	2																			BE	4	3							B6	4	2	✓	✓	-	-	-	-
LDY	M → Y (1)	A0	2	2	AC	4	3	A4	3	2													B4	4	2	BC	4	3													✓	✓	-	-	-	-
LSR	0 → [7 … 0] → C				4E	6	3	46	5	2	4A	2	1										56	6	2	5E	7	3													0	✓	✓	-	-	-
NOP	NO OPERATION													EA	2	1																									-	-	-	-	-	-
ORA	A∨M → A	09	2	2	0D	4	3	05	3	2							01	6	2	11	5	2	15	4	2	1D	4	3	19	4	3										✓	✓	-	-	-	-
PHA	A → M_S S-1 → S													48	3	1																									-	-	-	-	-	-
PHP	P → M_S S-1 → S													08	3	1																									-	-	-	-	-	-
PLA	S+1 → S M_S → A													68	4	1																									✓	✓	-	-	-	-
PLP	S+1 → S M_S → P													28	4	1																									(RESTORED)					
ROL	← [7 … 0] ← [C] ←				2E	6	3	26	5	2	2A	2	1										36	6	2	3E	7	3													✓	✓	✓	-	-	-
ROR	→ [C] → [7 … 0] →				6E	6	3	66	5	2	6A	2	1										76	6	2	7E	7	3													✓	✓	✓	-	-	-
RTI	(See Fig. 1) RTRN INT													40	6	1																									(RESTORED)					
RTS	(See Fig. 2) RTRN SUB													60	6	1																									-	-	-	-	-	-
SBC	A-M-$\bar{C}$ → A (1)	E9	2	2	ED	4	3	E5	3	2							E1	6	2	F1	5	2	F5	4	2	FD	4	3	F9	4	3										✓	✓	(3)	-	-	✓
SEC	1 → C													38	2	1																									-	-	1	-	-	-
SED	1 → D													F8	2	1																									-	-	-	-	1	-
SEI	1 → I													78	2	1																									-	-	-	1	-	-
STA	A → M				8D	4	3	85	3	2							81	6	2	91	6	2	95	4	2	9D	5	3	99	5	3										-	-	-	-	-	-
STX	X → M				8E	4	3	86	3	2																												96	4	2	-	-	-	-	-	-
STY	Y → M				8C	4	3	84	3	2													94	4	2																-	-	-	-	-	-
TAX	A → X													AA	2	1																									✓	✓	-	-	-	-
TAY	A → Y													A8	2	1																									✓	✓	-	-	-	-
TSX	S → X													BA	2	1																									✓	✓	-	-	-	-
TXA	X → A													8A	2	1																									✓	✓	-	-	-	-
TXS	X → S													9A	2	1																									-	-	-	-	-	-
TYA	Y → A													98	2	1																									✓	✓	-	-	-	-

(1) ADD 1 TO "N" IF PAGE BOUNDRY IS CROSSED.
(2) ADD 1 TO "N" IF BRANCH OCCURS TO SAME PAGE. ADD 2 TO "N" IF BRANCH OCCURS TO DIFFERENT PAGE.
(3) CARRY NOT = BORROW.
(4) IF IN DECIMAL MODE Z FLAG IS INVALID. ACCUMULATOR MUST BE CHECKED FOR ZERO RESULT.

X INDEX X
Y INDEX Y
A ACCUMULATOR
M MEMORY PER EFFECTIVE ADDRESS
M_S MEMORY PER STACK POINTER

\+ ADD
— SUBTRACT
∧ AND
∨ OR

⊻ EXCLUSIVE OR
✓ MODIFIED
\- NOT MODIFIED
M_7 MEMORY BIT 7
M_6 MEMORY BIT 6

N NO. CYCLES
\# NO. BYTES

OP-CODE TABLE

MSD \ LSD	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	LSD / MSD
0	BRK	ORA-IND, X				ORA-Z, Page	ASL-Z,PAGE		PHP	ORA-IMM	ASL-A			ORA ABS	ASL-ABS		0
1	BPL	ORA-IND, Y				ORA-Z, Page,X	ASL-Z,Page,X		CLC	ORA ABS,Y				ORA-ABS,X	ASL-ABS, X		1
2	JSR	AND-IND,X			BIT-Z,Page	AND-Z, Page	ROL-Z, Page		PLP	AND-IMM	ROL A		BIT ABS	AND ABS	ROL ABS		2
3	BMI	AND-IND,Y				AND-Z,Page,X	ROL-Z,Page,X		SEC	AND-ABS,Y				AND ABS,X	ROL ABS,X		3
4	RTI	EOR-IND,X				EOR-Z, Page	LSR-Z,Page		PHA	EOR-IMM	LSR-A		JMP-ABS	EOR ABS	LSR ABS		4
5	BVC	EOR-IND,Y				EOR-Z,Page,X	LSR-Z,Page,X		CLI	EOR ABS,Y				EOR ABS, X	LSR-ABS, X		5
6	RTS	ADC-IND, X				ADC-Z, Page			PLA	ADC-IMM			JMP IND	ADC-ABS			6
7	BVS	ADC-IND, Y				ADC-Z,PageX			SEI	ADC-ABS,Y				ADC-ABS, X			7
8		STA-IND, X			STY-Z, Page	STA-Z, Page	STX-Z, Page		DEY		TXA		STY-ABS	STA-ABS	STX-ABS		8
9	BCC	STA-IND, Y			STY-Z,Page,X	STA-Z,Page,X	STX-Z,Page,Y		TYA	STA-ABS,Y	TXS			STA-ABS, X			9
A	LDY-IMM	LDA-IND, X	LDX-IMM		LDY-Z,Page	LDA-Z, Page	LDX-Z, Page		TAY	LDA-IMM	TAX		LDY-ABS	LDA ABS	LDX-ABS		A
B	BCS	LDA-IND, Y			LDY-Z,PageX	LDA-Z,Page,X	LDX-Z,PageY		CLV	LDA-ABS, Y	TSX		LDY-ABS, X	LDA-ABS, X	LDX-ABS, Y		B
C	CPY-IMM	CMP-IND, X			CPY-Z,Page	CMP-Z, Page	DEC-Z, Page		INY	CMP-IMM	DEX		CPY ABS	CMP ABS	DEC ABS		C
D	BNE	CMP-IND, Y				CMP-Z,Page,X	DEC-Z,Page,X		CLD	CMP-ABS, Y				CMP-ABS, X	DEC-ABS, X		D
E	CPX-IMM	SBC-IND, X			CPX-Z, Page	SBC-Z, Page	INC-Z, Page		INX	SBC-IMM	NOP		CPX ABS	SBC-ABS	INC-ABS		E
F	BEQ	SBC-IND, Y				SBC-Z,Page,X	INC-Z,Page,X		SED	SBC ABS, Y				SBC-ABS, X	INC-ABS, X		F

IMM - IMMEDIATE ADDRESSING – THE OPERAND IS CONTAINED IN THE SECOND BYTE OF THE INSTRUCTION.

ABS - ABSOLUTE ADDRESSING – THE SECOND BYTE OF THE INSTRUCTION CONTAINS THE 8 LOW ORDER BITS OF THE EFFECTIVE ADDRESS. THE THIRD BYTE CONTAINS THE 8 HIGH ORDER BITS OF THE EFFECTIVE ADDRESS.

Z, PAGE- ZERO PAGE ADDRESSING – SECOND BYTE CONTAINS THE 8 LOW ORDER BITS OF THE EFFECTIVE ADDRESS. THE 8 HIGH ORDER BITS ARE ZERO.

A. - ACCUMULATOR - ONE BYTE INSTRUCTION OPERATING ON THE ACCUMULATOR.

Z, PAGE,X Z, PAGE,Y - ZERO PAGE INDEXED - THE SECOND BYTE OF THE INSTRUCTION IS ADDED TO THE INDEX (CARRY IS DROPPED) TO FORM THE LOW ORDER BYTE OF THE EA. THE HIGH ORDER BYTE OF THE EA IS ZEROS.

ABS,X ABS,Y ABSOLUTE INDEXED THE EFFECTIVE ADDRESS IS FORMED BY ADDING THE INDEX TO THE SECOND AND THIRD BYTE OF THE INSTRUCTION

(IND, X) - INDEXED INDIRECT – THE SECOND BYTE OF THE INSTRUCTION IS ADDED TO THE X-INDEX, DISCARDING THE CARRY. THE RESULTS POINTS TO A LOCATION ON PAGE ZERO WHICH CONTAINS THE 8 LOW ORDER BITS OF THE EA, THE NEXT BYTE CONTAINS THE 8 HIGH ORDER BITS.

(IND),Y - INDIRECT INDEXED – THE SECOND BYTE OF THE INSTRUCTION POINTS TO A LOCATION IN PAGE ZERO. THE CONTENTS OF THIS MEMORY LOCATION IS ADDED TO THE Y INDEX, THE RESULT BEING THE LOW ORDER EIGHT BITS OF THE EA. THE CARRY FROM THIS OPERATION IS ADDED TO THE CONTENTS OF THE NEXT PAGE ZERO LOCATION, THE RESULT BEING THE 8 HIGH ORDER BITS OF THE EA.

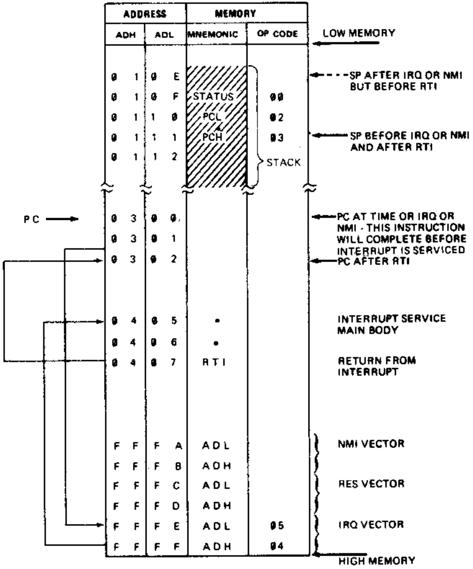


FIG. 1 IRQ, NMI, RTI, BRK OPERATION

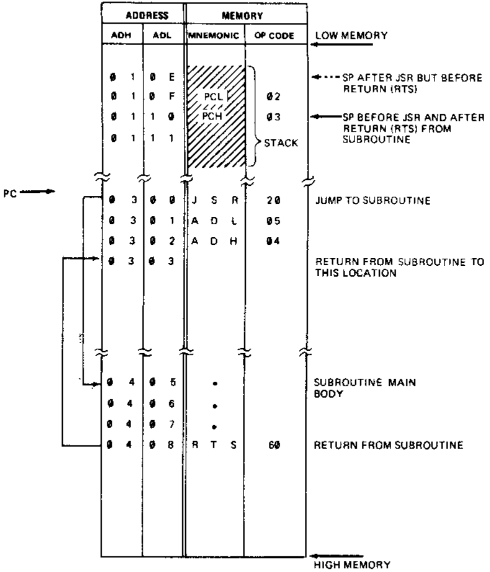


FIG. 2 JSR, RTS OPERATION

ASSEMBLER DIRECTIVES

- •OPT - IF USED MUST BE THE FIRST EXECUTABLE STATEMENT IN THE PROGRAM.
- •OPTIONS ARE:- (OPTIONS LISTED ARE THE DEFAULT VALUE. TURNED OFF BY (NO) PREFIX.)
  - COUNT (COU OR CNT) - LIST ALL INSTRUCTIONS AND THEIR USAGE.
  - NOGENERATE (NOG) - DO NOT GENERATE MORE THAN ONE LINE OF CODE FOR ASCII STRINGS.
  - XREF (XRE) - PRODUCE A CROSS-REFERENCE LIST IN THE SYMBOL TABLE.
  - ERRORS (ERR) - CREATE AN ERROR FILE.
  - MEMORY (MEM) - CREATE AN ASSEMBLER OBJECT OUTPUT FILE.
  - LIST (LIS) - PRODUCE A FULL ASSEMBLY LISTING.
- •BYTE - PRODUCES A SINGLE BYTE IN MEMORY EQUAL TO EACH OPERAND SPECIFIED.
- •WORD - PRODUCES TWO BYTES IN MEMORY EQUAL TO EACH OPERAND SPECIFIED.
- *= - DEFINES THE BEGINNING OF A NEW PROGRAM COUNTER SEQUENCE.
- •PAGE - ADVANCES THE LISTING TO THE TOP OF A NEW PAGE.
- •END - DEFINES THE END OF A SOURCE PROGRAM.

LABELS:

LABELS BEGIN IN COLUMN 1 AND ARE SEPARATED FROM THE INSTRUCTION BY AT LEAST ONE SPACE.
LABELS CAN BE UP TO 6 ALPHANUMENIC CHARACTERS LONG AND MUST BEGIN WITH AN ALPHA CHARACTER.
A, X, Y, S, AND P ARE RESERVED AND CANNOT BE USED AS LABELS.

LABEL = EXPRESSION CAN BE USED TO EQUATE LABELS TO INSTRUCTIONS.

LABEL * = * + N CAN BE USED TO RESERVE AREAS IN MEMORY

CHARACTERS USED AS SPECIAL PREFIXES:

•	INDICATES AN ASSEMBLER DIRECTIVE
#	SPECIFIES THE IMMEDIATE MODE OF ADDRESSING.
$	SPECIFIES A HEXADECIMAL CHARACTER.
@	SPECIFIES AN OCTAL NUMBER.
%	SPECIFIES A BINARY NUMBER.
'	SPECIFIES AN ASCII LITERAL CHARACTER.
( )	INDICATES INDIRECT ADDRESSING.
;	IN COLUMN 1 INDICATES A COMMENT.

ASCII CHARACTER SET (7-BIT CODE)

LSD \ MSD		Ø ØØØ	1 ØØ1	2 Ø1Ø	3 Ø11	4 1ØØ	5 1Ø1	6 11Ø	7 111
Ø	ØØØØ	NUL	DLE	SP	Ø	@	P		p
1	ØØØ1	SOH	DC1	!	1	A	Q	a	q
2	ØØ1Ø	STX	DC2	"	2	B	R	b	r
3	ØØ11	ETX	DC3	#	3	C	S	c	s
4	Ø1ØØ	EOT	DC4	$	4	D	T	d	t
5	Ø1Ø1	ENG	NAK	%	5	E	U	e	u
6	Ø11Ø	ACK	SYN	&	6	F	V	f	v
7	Ø111	BEL	ETB	'	7	G	W	g	w
8	1ØØØ	BS	CAN	(	8	H	X	h	x
9	1ØØ1	HT	EM	)	9	I	Y	i	y
A	1Ø1Ø	LF	SUB	*	:	J	Z	j	z
B	1Ø11	VT	ESC	+	;	K	[	k	
C	11ØØ	FF	FS	,	<	L	\	l	|
D	11Ø1	CR	GS	–	=	M	]	m	
E	111Ø	SO	RS	•	>	N	↑	n	~
F	1111	SI	VS	/	?	□	←	o	DEL